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Sound Design and Audiovisual Memory in the Instantiation of Digital Game Worlds: An Essay on *Iconophonics*

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Abstract: The article approaches the relationship between design, memory and technoculture through a conceptual discussion of audiovisual culture and the techno-aesthetic expressions of sound design in digital games. Hence, instead of describing it for its technical aspects, sound design is more broadly framed here as a technocultural practice deeply related to an audiovisual memory of media. The paper also presents the concept of iconophonics, by articulating this theoretical framework with an analysis of sound design features in five digital games: *Jazzpunk* (2014), *Rayman: Legends* (2011), *South Park: Stick of Truth* (2014), *Inside* (2016) and *Spec Ops: The Line* (2012).

Keywords: sound design, audiovisual memory, digital games, technoculture, media studies

1. Introduction

This article proposes a theoretical reflection on the role played by sound design of cultural artefacts in the mnemonic ways humans experience the world, arguing how the sounds of audiovisual media instantiate technocultural motifs of a given epoch. With this, it wishes to outline how sound design is simultaneously a product and driver of a wider cultural ambience. By emphasizing the relationship between design, memory, and technoculture, the essay suggests understanding digital game sound design, often discussed for its technical aspects, as a historically developed modern practice directly related to contemporary audiovisual ecology. With this theoretical framework in mind, the analysis of some contemporary digital games is carried out in light of the concept of *iconophonics* (Luersen, 2020). Iconophonics refers to sound punctuations that function as icons, allowing the audible dimension of games to establish particular models of their audiovisual worlds which are still relatable to our quotidian and affective experience with sounds.

Iconophonic constructs shed light on the non-explicit and continuously renewed relationships between design, technoculture, and memory. This approach holds that technoculture produces sounds that digital games explore creatively, expressing variations of the acoustic ecology of the world at a given moment. Conversely, the techno-aesthetics of computer-simulated worlds also disseminate culturally, intertwining technical and cultural dimensions like in a feedback loop (Shaw, 2008). The article demonstrates, furthermore, how the sound modelling for contemporary

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technical media such as digital games develop from an important relationship with audiovisual memory. The sound design of digital games shares a common semantics with the aural expressions from other media like cinema, television, radio, and more broadly, with a memory of contemporary audiovisual culture. This nourishment of sound design by images of different natures relaxes our quotidian experience in a highly technically mediated world and nuances our relationship with cultural artefacts like computer games.

2. Audiovisual design as product and driver of cultural codes

The worlds imagined by games are permeated by the aesthetic experience of a technocultural environment that surrounds them. The flow of newly encoded sounds and their recodifications can be understood as a constant operation of harmonization and disharmonization within technoculture, representing a *neguentropic* endeavour that shapes the human experience of the world: Vilém Flusser suggests that the motivations to communicate are artificial. He proposes that humans relate to nature through repeated attempts to deceive it, by challenging the entropic principle through the production and accumulation of information (Flusser, 2015, p. 269). In his lectures in Bochum, Flusser states that human communication is first and foremost a counter-natural means to ‘‘storage, process, and transmit acquired information’’ (Flusser, 2015, p. 33).

According to Flusser, humans are artists capable of imagining and designing worlds with formal order. The advent of computing has complexified human communication, as we make a highly codified world of numerical abstractions apparent, filling imagined forms with matter through technical equipment embedded with scientific concepts. With this, Flusser observes that codes become *second nature*, leading mankind to forget their artificiality and how we codify the world through invented devices.

Through this lens, one can understand that culture and the human experience of the world have always been mediated by the relationship between individuals and techniques, as communicated through the materialities of communication. Each different technique introduces new layers of code. The anthropologist Massimo Canevacci (1990, p. 22) uses the term *videoscape* to refer to the panorama of technically reproducible audiovisual communication codes that influence human habitual perception by introducing electronic media codes into everyday life. What one knows of the habitual visual and aural landscape through their perception, by looking and listening, is pre-mediated by technoculture. For example, how many of us have visited Patagonia, climbed a mountain, or travelled in a hot-air balloon? Even if we have not been through any of these experiences, though, most of us possess at least a mental image of them. That is, even if we have not experienced certain places or situations, we possess mental images of them, which leave a mark on our overall experience of the world. If we ever have the opportunity to realise any of these situations, the experience will be tinted by these pre-existing mental images (Cardoso, 2016, p. 60).

But this does not refer only to these highly picturesque images, but also applies to trivial everyday tasks. Thus, to stay with the terminology used by Canevacci, the codes of the videoscape come to be assumed as living nature and are projected in the visual culture and habits of human experience.

As such, the codes of the audiovisual media landscape also impact everyday experiences with sounds. For instance, television and radio (where studio sound is recorded in a dampened environment, without reverberation, and foregrounding the voice) created models of a so-called ‘natural’ orality, which apply more-than-natural intelligibility to human speech, influencing the way people speak in collective spaces.

One does not need to look only to "media-trained" speech professionals, but can also notice, as Michel Chion (1999, p. 44) does, how people grow accustomed to specific prosody, articulation, accent and speech speed inspired by mass media, and how expectations are set for an "enabled speaker", especially when people communicate in public spaces (which are themselves noisy and reverberant, like classrooms, for example).

Several sounds are taken for granted and presumed to be well-known based on cultural experiences mediated by sound design constructs. Nonetheless, the technically-mediated audiovisual memory can be misleading. In the field of forensic audio analysis (Maher, 2009), for example, where recordings serve as potential criminal evidence, technicians often have to explain to the jury that the sound of a gunshot in a recording is different from the sound they have heard previously and that they hold as an acoustic referent. The sound of gunshots in Triple-A games[2], TV shows and blockbuster movies usually have their duration sustained due to the post-production application of heavy signal processing effects, especially reverberation and echo, which produces a more impactful sound effect.

Holger Schulze (2019, p. 16) proposes that sound design is analogous to magic tricks. For him, the sound design practices of audiovisual media can be related to the sound artifices performed by folk artists in travelling bazaars and fairs of curiosities from the turn of the nineteenth to the twentieth century, and should be linked to the effects evoked by spectacles of illusionism. However, Schulze also notes that with electronic and digital media, sound tricks have lost their extravagant character and, with the permeation of sound design into everyday routines, they have gradually become habitual. One can think, for example, about the imaginary references that people have about the sound of extinct species of animals and entirely outdated or obsolete machines. Contemporarily, we have mental images of how dinosaurs of different sizes could have sounded and can imagine the different sounds produced by a steam locomotive. Mental images of different environments and eras proliferate, and these artificial acoustic ecologies can be heard in an applied form in digital games simulating major historical events from the past (even though these sounds may not have any factual correspondence with their primary sources). In the same sense, the imaginary sounds of science fiction-themed audiovisual media also tint how present-day culture imagines the acoustic ecology from times to come.

3. Sound design and habituation: the creative dimension of acoustic memory

All these mental images of sounds, including those that refer to contemporary events heard in everyday life, are mediated and nuanced by other images and techniques. In the communicative processing of a code, the medium is never just an objective and transparent transportation mechanism, for it endemically affects what is conveyed. This is especially significant to the technically-mediated acoustic worlds constructed since the principles of recording technology, as Jonathan Sterne points out:

> When one traces recordings back to their so-called sources, one finds the intersection of cultural forces that made initial and subsequent moments of reproducibility desirable

[2] *Triple-A*, or *AAA*, refers to a "quality standard" in the games industry, agreed upon through a series of meetings between game developers and publishers in the 1990s (Demaria & Wilson, 2002). They often describe high-budget video games produced by major game development companies or publishers and feature extensive marketing campaigns. Triple-A games are expected to target a wide audience.

> and possible. There was no "unified whole" or idealized performance from which the sound in the recording was then alienated. To whom we attribute the possibility and the desire to record, or listen is entirely context dependent. Recording is a form of exteriority: it does not preserve a preexisting sonic event as it happens so much as it creates and organizes sonic events for the possibility of preservation and repetition. Recording is, therefore, discontinuous with the "live" events that it is sometimes said to represent (although there are links, of course). Like the body embalmed, recorded sound continues to be able to have a social presence or significance precisely because its interior composition is transformed in the very process of recording (Sterne, 2003, pp. 332-333).

The form of exteriority mentioned by Sterne when referring to sound recording is further radicalised with the possibility of synthesising sounds with computers. And even if with sound synthesis it is eventually possible to accurately reconstruct certain sound objects not accessible as "sources" in the present, the acculturated nature of listening can inevitably only perceive these with present-day ears. As Peter Krapp (2011, p. 71) puts it, human hearing is always influenced by both cultural and physiological filters. Mere habit plays a significant role in the process of acculturation to the sound icons constructed by sound design, and this understanding helps to highlight how the boundaries of culture and nature are not always that clear in how the audible dimension of digital media is perceived.

The possibility of sharing the imaginaries that permeate habituated perception allows one to establish a notion of everyday reality. It is what facilitates the establishment of a common basis to talk "about something", for instance. As Cornelius Castoriadis (1986, p. 13) explains, the imaginary incessantly creates figures and forms, projecting certain expectations and collective representations onto matter. This emphasizes the continuous feedback loop between codes of different natures and the production of new information in audiovisual communicative processes, which evolve from previous experiences.

In the context of design, especially in the rhetorics concerning digital games, there is a common sense notion that aesthetics are about novelty, even if they rely heavily on mimicry, realism, and hyper-realism. Therefore, it might come as a surprise that contemporary game sound design draws references from pre-coded images of technoculture even more often than in the pre-digital media age. The technical condition of creating digitally without necessarily addressing a "source" referent result, more often than not, in a strong representational emphasis. Therefore, it is wise to overcome a straight binary opposition between tradition and innovation when it comes to examining the technical and aesthetic aspects of digital games.

In game sound design, this mixed process is evident in the persistence of anthropomorphical representations and the construction of voices to represent humanoid traits for characters. The construction of voices relies on naturalized sound codes and techniques that are easily recognizable in the audiovisual aesthetics of other media. This includes using human voice recordings and dubbing in post-production (widely used in animated movies), or the use of vocal gestures imitating the articulations of radio and television voiceovers, even when voices do not perform a narration function. Additionally, when games represent computer voices or non-human characters by imitating computer-generated tones, they often use the voices of human actors as a basis, as can be noticed in games like *Portal 2* (2011) and *Bioshock* (2007). These sonic expressions, even when seeking mimicry, result in hybrid aesthetics, where sound languages from other means intersect with computer-based constructions. This produces differences in sounds that aim for similarity, which eventually become new audiovisual codes that reinsert themselves in the feedback loop of audiovisual culture and acoustic ecology.

The sound design of games–as inclined to innovation as they may be–operate under dialectics of similarity and dissimilarity, under questions of matter, imitation and cognoscibility, as previous media and art did before. Even if representation is not their primary technical mode of operation (Weissberg, 1999), these aesthetic relations contribute to the modelling of digital game worlds, sometimes originating from mundane sound recording and human dubbing processes, while at other times utilizing digital signal processing (PSD) effects. Hence, the affinity that these codes establish with a habitus leads us back to Flusser's concern about the information of second-nature worlds, as their interpretation play a decisive role in how habituation processes absorb cultural tropes that emerge when new technical artifices are introduced. Elizabeth Grosz further refines the role of habituation in this cultural dynamic, defining it as a "creative capacity" (Grosz, 2013, p. 219) that opens up the possibility of stability in a universe in which change is fundamental. Habit would be the way through which lived regularities are organized in moments of cohesion and repetition, in a universe where the past accumulates with unrelenting force and in which nothing truly repeats: "Habit not only anchors a site of regularity in a universe of perpetual change; it initiates change in the apparently unchanging, it opens up the possibility of understanding the very force of temporality itself, the force that adheres the past to the present and orients both to the possibilities of action in the future." (Grosz, 2013, p. 233).

Therefore, habit is a characteristic of cultivated nature, one that persists as a mnemonic technique. Individual conscious action is preceded by unconscious habits, where the irrational finds a comfortable space for creation. Habit enables communication, serving as a means to structure organic pre-social, pre-individual forms of relation, as well as to organize additional, acquired dispositions. It connects individuals and environments, while also educating and orienting their dispositions towards or against it. Wendy Chun (2016) suggests that *habitus* endures as a societal residue in the collective groupings of contemporary neoliberalism, when the possibility of a proper society, organized around communal bonds and relations, may no longer even exist. Moreover, habits change subtly, nurturing unexpected variations from one state to another. Through practice, as new cultural codes emerge from these background shifts, habit inevitably leads to situations of crisis (Chun, 2016), a necessary condition for the cycles of cultural updating.

A similar logic operates in the composition and engendering of sound design within a broader, persisting acoustic ecology, in which acoustic cultural codes gradually take on a second-nature condition. Contemporary sound design for games, apps, platforms, and other digital services and devices draws upon and remodels these previous aesthetic conventions, musical repertoires, and diverse sonic materials.

Designers can construct certain situations and stimulate listeners to recall specific moments through sound, leveraging this acoustic memory. In this sense, memory allows sound designers to simulate spaces and distances, suggest the action of contact between objects (such as rubbing, rebounding, tearing, breaking, absorbing), indicate material qualities (rough or smooth, hard or soft, solid or hollow, metal or wood), or allude to a volume, a mass, a dimension. *Phonomnesis*, that is, the ability to "imagine or remember a sound without necessarily hearing it" (Augoyard & Torgue, 2006, p. 60), plays a vital role in design practices as it implies the recollection of a referent, a unit to be repeated, not necessarily with any fidelity. As in many cases with human memory, the process of recollection takes precedence over the referent, updating the recalled mental image and making the act of repetition itself as significant as that which is remembered.

Furthermore, it is important to consider that, while modelled sounds depend on certain acoustic qualities nurtured and repeated over time, they are also influenced by conditions of the environment where they are reproduced. This means that exogenous factors can completely modify the associations made during acoustic recollection. For example, in the nineteenth century the melody of Ludwig van Beethoven's *Bagatelle No. 25 in A minor* (popularly known as *Für Elise*) may have been a cultural sign of "noble" courtship, but its meaning is radically different when heard on a telephone line queue. In Brazil, when the initial notes of the solo piano piece were used in the jingle tune for the country's main gas delivery company in the 1990s, hearing the melody on a Monday morning conveyed a much more proverbial cultural meaning.

As in such cases, any emergent expressive sound technology, while initially showing up as a new artifice in the inventory of acoustic codes, is bound to be reinterpreted within the broader acoustic ecology of adjacent media, where it is negotiated with habituated audiovisual codes. These codes are experienced in relation to mental images and sounds culturally transmitted, preserved, repeated. As widely distributed media in contemporary times, digital games participate in the recodification of cultural habits. Through their sonic expressions, they stretch the boundaries of what still sounds inorganic to contemporary culture within a more comprehensive audiovisual ecology. In effect, games also contribute to the construction of audiovisual memory by leveraging this quasi-anthropological aspect of sound that participates in the attribution of meaning to technically-built audiovisual worlds.

The game *Pong* (1972), often cited as the first coin-operated arcade game, can help illustrate this argument. The machine was made to be accommodated in waiting rooms and bars, and to make it commercial it was necessary to invent both "a start, to allow the insertion of a coin," and "an end, to require the insertion of additional coins" (Pias, 2004, p.128). Due to the simple technical features of the arcades, the sound objects constructed in the early games needed to have a low sonic complexity, while also requiring the noises to be sufficiently varied to represent the different actions in the game (Collins, 2012). This gave rise to the distinctive timbres produced by square waves, such as the onomatopoeic *beeps* heard when the ball is rebounded in Pong, for example. The "ball" itself was represented by a square, as it was simpler to depict it without rounded corners in a pixel map.[3] The timbres were limited by the types of oscillators in the hardware, which led to simply punctuating the gestures and actions of the players with abstract noises.

However, even such abstract sounds as the *beeps* and *bloops*, among other non-figurative sound signals of the early generations of arcades and consoles, ended up evolving as a non-stated orientation for the development of sounds in digital games. Even in contemporary games with hyper-realistic aesthetics, the imprint of these early-generation sounds still shows up, whether during gameplay or, more frequently, in configuration menus. Confirming actions in menus are typically constructed with clearer timbres, while cancelling actions are punctuated by more opaque sounds. These sonic properties punctuate actions, attributing simple symbolic meaning to the movements made by the player in the controller.

[3] Nolan Bushnell, founder of Atari, stated in an interview conceded to the Los Angeles Times: "At the time, it was very difficult to operate technically. We didn't create a square ball for Pong because we thought it would look cool. We did it because it was all we could do." One can also understand how this relates to the development of sound for games. Available at: https://shorturl.at/aVpgN (last accessed, May 28, 2024).

It is also worth mentioning that an important characteristic of sound design in more complex digital games is its dynamic arragement[4], where the behaviour of sounds, their processing, and articulations are based on preset, responsive audio systems. In this case, certain sound triggers are set to micromanage the acoustic events during gameplay from a larger set of variables, which can still include the gestures and actions of players. Therefore, the sounds seem to break down the actions performed in the game world into micro-behaviours, with the audio system operating like a live score, adjusted to real-time performance. In this sense, the nature of audiovisual synchrony in digital games is best understood from the perspective of machine agency, which subsumes the performance of objects, the simulation of space, the movements in the environment, the behaviour of music, and other dynamics of sound articulation written for the game, to be executed and managed by the computer.

4. Iconophonics: the role of acoustic memory in the sound design of audiovisual worlds

The convergence between technically reproduced sounds and the production of entirely artificial digital environments in the poetics of space in digital games represents well how the technical developments previously achieved in research laboratories linked to the music industry (especially the development of stereophonic sound) interlapse with the ability to process and store data through computers in the last decades. The sense of immersion often used to describe how sound is experienced in audiovisual media is often associated with the spatial characteristics that constitute it, especially concerning the attempts to model the navigational environments of digital games. In such an experience, the relationship between the modelled spaces and *time* is essential.

Sounds, decorating the temporal experience of gameplay, might better support the description of this *sensation* of immersion as a bodily habit. Sounds not only connect gameworld spaces to exogenous reality in an imaginary fashion, but also merge different instances and sections of a game with each other, punctuating gameplays with an internal thread of *leitmotifs*. These repeated elements imaginatively signify the space of the game as a world in itself, making the experiences of a game recognizable and memorable, constituting also a listening habit of a particular audiovisual world for players. Just as game graphics, sound design plays a crucial role in making such a world seem like a unique entity, a concrete whole, as it instantiates a simulated environment with distinct sonic assets and idiosyncratic expressions.

Certain modelled sounds become iconic auditory references of a game environment and, like knots in a thread, make players connect their mnemonic experience with its iterative acoustic ecology, serving as entry and exit markers of this imaginary world. These elements of sound design are referred to here as *iconophonics*. Iconophonic constructs are instrumental in instantiating a game environment, offering a concrete and consistent acoustic environment to be revisited oftentimes during the widely fragmented and dispersed experience of gaming, as hours of gameplay are often very fractioned by players. Iconophonics describe recurring, memorable sounds that provide a sense of coherence to the experience of each game. In the following pages, the article

[4] As this article does not mean and cannot exhaust such a broad topic as the historical influences on the sound design of digital games, the aspect of algorithmic management of the audible dimension of games is only briefly mentioned here. This is an important issue, but as this article focuses on the question of iconophonies, interested readers can be directed to a previously published work (Luersen, 2020), where the automation of audiovisual synchronization in digital games is approached in more detail.

analyses the audible dimension of selected contemporary digital games in order to exemplify different types of iconophonic constructions.

For instance, in the game *Jazzpunk* (2014), right from the start, players find themselves in a closed corridor, where they hear the sound of pressed keys and typing in a typewriter being continuously used. As they progress along the corridor, visual signs and objects indicate that they are entering an office. The source of the sound is revealed when they enter the room and see a secretary typing non-stop. Even when the game is paused or when interacting with the character, the sound of the keys persists, creating a *sound gag* of uninterrupted typing sounds. The character voices are deliberately barely intelligible–they sound distorted and altered by audio filters. As players interact with objects in the environment, they hear out-of-tune string sounds and other histrionic noises that mock the manipulation of any object. A door opens with a rapid, shrill onomatopoeia, reminiscent of the sound design of old TV show cartoons. These and many other sonic gags are gradually introduced and accompany the player throughout the game, evoking some acoustic conventions of genres such as parody and satire. Most objects in the environment of the game respond to the actions performed by players with a jocose, playful tone. The constant iteration of these iconic sound elements constitutes the particular experience of this gameworld.

Throughout the experience, these iconophonic constructs form a fabric of sonic patches that exhibit their meaning in the cohesion they form during the playthrough of Jazzpunk. On a second listen, the exaggerated and dissonant sounds blend to create a comic atmosphere that becomes habitual due to the successive repetition of sonic gags. As a result, by the time players advance further in the game, they expect every object they touch to provoke some absurd sonic response. The repetition of these sounds conditions the expectations of players, establishing an internal habit of the game experience itself. Sound design contributes to producing an internally coherent ambience, which will be extensively recollected each time the player returns to the game, at various points throughout its duration, as is typical in the dispersive experience of a digital game.

This seemingly coherent ambience is partially built by iconophonics, which are set to reconstruct a mental image of specific objects by intertwining the presentation of sonic events in the gameworld. Listening to these sounds evoke a memory of previous game experiences, attaching the mental image of the environment to that world each time the player returns to it.

The game *Rayman Legends* (2013) also provides an interesting example of how iconophonics operate by overlaying exogenous memories and iterative game sounds. Once players join the game, the gameworld immediately evokes an infantilized atmosphere built, which is reiterated by the sound design throughout the experience. Character exclamations, tunes based on short, catchy melodies in major keys, the histrionic sonic punctuations of clapping, booing, slapped kisses, and a range of other sound design elements, all create an acoustic counterpart to the colourful, glossy visual design of the game.

Brightness is vital in Rayman, as the acoustic ecology of the game deliberately operates by reconstructing sounds associated with the memorabilia of a child's party and play. When the protagonist takes damage from an opponent, for instance, the player clearly hears the growing tension of a balloon's skin being stretched. Other times, they hear the squeak of a full balloon's skin being scratched, the huff of a bladder rapidly emptying itself or slowly squeaking until it crumbles. The sound of balloons popping can also be heard in several places, alongside a series of other verbal onomatopoeia, interjections, claps and whistles that connect the various levels and scenarios during the experience of the full game.

Hence, the material indexes and textures recalled in the modelling of the sound cues for this game connect it semantically to a familiar infantilized universe that exists external to the game. Internally, these textures and elocutions create a *cuddly* environment, despite the gameplay portraying actions like crushing smaller beings, blowing them up, slapping them, or throwing them against each other–all very traditional elements of platform games. These sonic motifs seem to convey an acoustic analogous to what Gisele Beiguelman (2011) calls the visual expressions of *cute capitalism*, an audiovisual aesthetic trope easily adaptable to the needs of a neoliberal economy of *like* buttons. Through such an "iconophonic cuteness," the sound design of these worlds constructs celebratory environments that mitigate the tropes of conflict and dissent–which explains why such aesthetics are also so widely used in contemporary sound branding, especially in the sonic punctuation provided for messaging apps and social media platforms. In their acoustic dimension, such environments quickly recall the exclamation sounds of instant communication platforms of several different generations (*ICQ*, *Messenger*, *WhatsApp*). Even if a Ubisoft title, games like Rayman can be strongly associated, at least in their sound aspects, with the celebrated *kawaii* aesthetics that often guides the design projects of developers like Nintendo (*Kirby's Dream Land, Super Mario World, Animal Crossing*). Therefore, on the one hand, iconophonics links the various different parts of the game as an internally coherent whole, while on the other it weaves the acoustic ecology of the title to identifiable imaginary, external tropes. This experience can be described, phenomenologically, as producing a surrounding environment which is both self-centred and shared, which connects the particular gameworld to prosaic events of the player's memory. The game as an imaginary experience is enabled precisely through the communicative process happening between these two domains, with sound design serving as an instrument of constant reification and instantiation of this relationship, one that extends a memory of sounds into the aesthetic experience of gameplay.

In the game *South Park: Stick of Truth* (2014), an additional type of iconophonics permeates the sound design, and points the player towards their sonic experience with previously established audiovisual worlds. Stick of Truth includes various audiovisual snippets that naturally refer, quite directly, to the South Park animated TV series. The game reuses materials from the series, in which sound design is treated with significant aesthetic concern. Several experiments are made with the game form, though, using satire to stretch some of its formal properties. The focus of the game's acoustic ecology, though, is in how it repurposes elements of the series' imagery in order to provide thematic consistency to the game version of the South Park universe.

Several musical elements of the series naturally appear in the game, such as vocal choirs with pitch-shifted voices, pastiches of well-known music tracks, and the contrasting mix of crystal clear instrumental tones with grotesque screechy voices. Examples are the comic choirs in the church area and the randomly triggered soundtracks of movie trailers that can be heard at the movie theatre. Moreover, background music in the game often also serves a more prominent role than merely supplying a backing track. Specifically, the game features a tracklist of the South Park show that plays at various moments throughout the game. This ambient tracklist helps in supporting the seemingly "authentic" adaptation of the game's aesthetics with respect to the *ethos* of the TV show, while creating a congruent acoustic ecology for the different spaces of the game experience at the same time. The ambient tracklist, playing themes from the series in random order and with several different audio filters applied, can be heard with varying intensities in most shops and public buildings in the main scenario of the game, the city of South Park. This is a comic interaction of the game with the concept of *muzak*.[5] The content of the songs changes each time the player enters a building to correspond with changes in characters and game situations. As a result, the tracklist itself functions as an iconophonic motif throughout the gameplay. When entering the townhouses of South Park, one can hear the sound of television sets being turned on. Most of the time, since the TV is depicted with its back to the image, the representation of a program (based on events in the series) is implied, as a diegetic sound that plays in the background as players explore the rooms. Therefore, the musical motifs of the series serve as a central part of the acoustic ecology of the game, creating the impression that players are playfully engaging with the TV show itself.

When analyzing this sonic construct, one particular characteristic of digital games comes forward, which is the dynamic micro-adaptation of sound or music to game states. Other sorts of short sonic motifs and micro-compositions, such as piano phrases, glockenspiel micro-tunes, and the sound of vibrating or plucked strings, react to game events and player actions. Therefore, iconophonics in games can operate not only to habituate players to the gameworld or to evoke intertextual references but can also detach music from its usual background function. The musical themes in Stick of Truth, in particular, fulfil this particular function as mnemonic triggers that make acoustic tropes of the TV show overlap game events, framing gaming situations *as if* players were directly playing the series.

In the game *Inside* (2016), the iconophonic element that is reinstated at different points of the gameplay and weaves the audible dimension of the gameworld together is, intriguingly enough, an acoustic property. Several different objects and sonic events of the game are modelled as low and deep rumbles that gradually invade the soundscape at key points of the playthrough. These are vivid noises that often endow the graphic images of engines, dams, and bombs with an animistic aspect. These sonic expressions are not necessarily figurative, and more often than not they convey a psychological effect to gaming situations. More than a representational function, these sonic expressions allure an intensive bodily response, as they give the impression of "filling in" the gameworld space with sound. It is remarkable in this game, therefore, how various situations of the game are connected by this mnemonically evoked low-tone trigger, which significantly tints the atmosphere of the game. While the protagonist is usually visually positioned at the centre of the screen, the grandiloquent and animistic sound design seems to make the main character smaller, in a diametrically inverted scale. All incidental noises caused by the player during these game situations are muffled by the bass drones that periodically bathe the atmosphere with a dense and heavy sound mass. Hence, an acoustic property becomes the main iconophonic element in the world of Inside: the sporadic low timbres, which are periodically evoked to envelop the atmospheric audiovisual experience of the game.

Finally, the main iconophonic element in the game *Spec Ops: The Line* (2012) follows a different trajectory, constantly presenting and sonically reshaping a representational element: the wind. As a sonic event, the wind in the game seems to convey several mental images through a series of behaviours which are attributed to it. It also suggests moods and conveys sensations through its association with the desert

[5] Also known as *elevator music*, Muzak is a generic term used to refer to a style of smooth arrangements of popular songs prepared to play in shopping centers, department stores, malls, and telephone systems (while the listener is on hold). The term was coined by the *Muzak Corporation*, a company still active in this business.

landscape in the game. At the very beginning of the playthrough, the first sounds of wind are experienced by players. The first audio cues signal a (very) sharp flaring of the USA flag, which is seen inverted and torn, centred on the screen during a cutscene.[6] After a fade-out, the gameplay starts, and the previous sounds cease, giving way to a sonic texture composed of three elements: a dialogue between the protagonist and a secondary character, the sound of footsteps in the sand, and a dense layer of ambient noise, which corresponds to the wind blowing irregularly. This wind cue is composed of short intermittent noises and abrupt high-pitched hisses, along with small granular effects in the whole soundscape, which are meant to simulate the grains of sand soaring through the landscape, frictioning against different surfaces. Throughout the gameplay, as new gaming situations and events unravel, wind sounds reappear successively, albeit with significant modulations at each time.

The variations in the modelling of windstorm sounds also instantiate an animistic setting for the gameplay. At each time, the modulations of this iconophonic element punctuate mood changes in the atmosphere and storyline. In battle situations, there is often an intensification of the windstorm, and the sounds of grainy sand frictions are temporarily amplified, as if the sound mass was enunciating the very intensity of the conflicts. Such sonic expressions seem to corroborate the idea that certain natural motifs, such as the wind, have allegorical and archetypal qualities. Without limiting themselves to a specific medium or context, these archetypical images are recollected in the sound design due to certain mnemonic qualities, which also extend far beyond their instrumental use in a game.

Due to its archetypical character, the wind has already been the subject of many literary descriptions, as Raymond Murray Schafer (1997, p. 42) reminds us. Such descriptions often sought to recall the memory of the wind to evoke moods and suggest psychological traits, as in the following excerpt from Victor Hugo's *Toilers of the Sea* (1866):

> The vast commotion of those solitudes has its gamut, a terrible crescendo. There are the gust, the squall, the storm, the gale, the tempest, the whirlwind, the waterspout–the seven chords of the lyre of the winds, the seven notes of the firmament. (...) The winds rush, fly, swoop down, dwindle away, commence again; hover above, whistle, roar, and smile; they are frenzied, wanton, unbridled, or sinking at ease upon the raging waves. Their howlings have a harmony of their own. They make all the heavens sonorous. They blow in the cloud as in a trumpet; they sing through the infinite space with the mingled tones of clarions, horns, bugles, and trumpets–a sort of Promethean fanfare. Such was the music of ancient Pan. Their harmonies are terrible. They have a colossal joy in the darkness. They drive and disperse great ships. Night and day, in all seasons, from the tropics to the pole, there is no truce; sounding their fatal trumpet through the tangled thickets of the clouds and waves, they pursue the grim chase of vessels in distress. They have their packs of bloodhounds, and take their pleasure, setting them to bark among the rocks and billows. They huddle the clouds together and drive them diverse. They mould and knead the supple waters as with a million hands (Hugo, 1866, pp. 300-301).

Moreover, as if a plastic, material model, tropes as the wind allow for a long range of possible acoustic variations. The white noises generated by TVs or radios out of tune, for example, sonically remind us of the wind due to their high level of abstraction, as broad-spectrum acoustic events. This is also why in early consoles, the white noise

[6] A cutscene is a non-interactive video sequence in a video game used for storytelling or showcasing important events. Players have no control during cutscenes and watch pre-rendered or scripted scenes unfold more often to advance the narrative of the game.

channel of sound cards and audio processing boards was very often utilized to build background, environmental elements, such as waves on the sea or the wind blowing in a forest. In general, these types of sounds are dominant in the audio spectrum, adding a significant bodily quality to game events. The abstract acoustic character of the wind also allows sound designers to imprint different qualities on the gameworld environments by operating micro-variations to the same sound cue at different points of the gameplay.

Such variability in the sound design of the wind becomes even clearer in a gaming situation referring to an intense battle near the end of the game. The soundscape of the game gets filled with variations of gunshots, explosions, and screams, which add to the background music to produce a cacophony that lingers for a considerable amount of time. A sudden change occurs as the last opponent is shot: the music is interrupted, and immediately, the player begins to hear the wind blowing in a crescendo. After all the cacophony, what remains is the gale and the ruins of an abandoned city. No one else is seen in the environment, and there are no more sonic punctuations of player actions. The abrupt silence evokes a latent state of abandonment in the middle of the desert. The sound design supports in dramatizing audiovisual situations such as this. The game wishes to convey the sense of a gradual loss of meaningfulness consummated over the course of a war. The irregular hissing and the granularities of the sand particles form a leitmotif, as throughout the game the acoustic punctuations of wind gushes cease to seem an amorphous mass of sound and gradually start to evoke an animistic sense, modulating the atmosphere of the game in kinship with previously existing audiovisual, aesthetic repertoires.

5. Concluding remarks

By generating congruence between audiovisual events of the endogenous environments of games, sound design contributes to establishing habits of navigation and acclimatization in each of the modelled worlds. These are sounds continuously recalled to punctuate certain gaming situations, permeating with mnemonic substance the experience of each simulation world. One can notice that, at first hand, the modelling of specific sounds can work in the sense of building a conductive thread that can further connect the many iterations of players within the gameworld.

Sound design itself produces a memory of the game. The audiovisual world can be properly recognized as a site for exploration, a space of simulation that is experienced phenomenologically as a “being and acting” in the environment. This situation updates and produces further technocultural motifs to be recollected by a cumulative audiovisual memory increasingly permeated by the aesthetics of digital games.

The prevalence and transmission of sound codes are tacitly perceived and absorbed by the cultural dynamics of acoustic communication. In light of this, one can reconsider the anthropotechnical aspect of sound design practices: for besides such codes being culturally and historically produced, their techno-aesthetic construction is intimately linked to their cycles of transmission and consumption in a given time. Their shaping is, therefore, a task that design constantly remakes, updating and implementing cultural variations in the artefacts it elaborates.

The jocose sound world of Jazzpunk, the reminiscent tracklist of South Park, the childish memorabilia in Rayman, the deep timbres in Inside, and the expressions of the wind in Spec Ops are some examples that show, nonetheless, more than a merely imitative role. In the effective construction of sound worlds in digital games the iconophonies, iconic sound models that connect the various passages and ellipses of the gameplay, instantiate aesthetic traces that connect players to the atmosphere of that world in particular. But they also link the game worlds to a wider cultural memory of sounds. That provides the conditions so that sound design, in its function of producing simulated world models, can feed with a whole psychic life the strange experience of manipulating plastic controls while looking at a luminous screen in domestic or portable devices.

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